it would spin twice as fast as a photon. None of the experiments had ever found such a particle, so this appeared to be among the erroneous predictions made by string theory.

Schwarz and his collaborator Joël Scherk puzzled over this case of a missing particle, until in a magnificent leap they made a connection to a completely different problem. Although no one had been able to combine general relativity and quantum mechanics, physicists had determined certain features that would emerge from any successful union. And, as indicated in Chapter 9, one feature they found was that just as the electromagnetic force is transmitted microscopically by photons, the gravitational force should be microscopically transmitted by another class of particles, gravitons (the most elementary, quantum bundles of gravity). Although gravitons have yet to be detected experimentally, the theoretical analyses all agreed that gravitons must have two properties: they must be massless and have spin-two. For Schwarz and Scherk this rang a loud bell—these were just the properties of the rogue particle predicted by string theory—and inspired them to make a bold move, one that would transform a failing of string theory into a striking success.

They proposed that string theory shouldn't be thought of as a quantum mechanical theory of the strong nuclear force. They argued that even though the theory had been discovered in an attempt to understand the strong force, it was actually the solution to a different problem. It was actually the first ever quantum mechanical theory of the *gravitational* force. They claimed that the massless spin-two particle predicted by string theory was the graviton, and that the equations of string theory necessarily embodied a quantum mechanical description of gravity.

Schwarz and Scherk published their proposal in 1974 and expected a major reaction from the physics community. Instead, their work was ignored. In retrospect, it's not hard to understand why. It seemed to some that the string concept had become a theory in search of an application. After the attempt to use string theory to explain the strong nuclear force had failed, it seemed as though its proponents wouldn't accept defeat and, instead, were flat out determined to find relevance for the theory elsewhere. Fuel was added to this view's fire when it became clear that Schwarz and Scherk needed to change the size of strings in their theory